

Amortized analysis using potential function

$$D_0 \xrightarrow{o_1} D_1 \xrightarrow{o_2} D_2 \xrightarrow{o_3} \dots D_i \dots D_n$$

$\phi : D_i \rightarrow \mathbb{R}$ potential function

$$\phi(D_i) = \phi_i \quad \Delta_i = \phi_{i+1} - \phi_i$$

$$\text{Amortized cost}_i = \text{Actual cost}_i + \Delta_i$$

Taking sum $\sum_{i=1}^n \text{Amortized cost} = \sum_i \text{Actual cost} + \sum_i \Delta_i$

$$\begin{aligned} \Rightarrow \text{Total amortized cost} &= \text{Total Cost} + \sum_{i=1}^n \phi_i - \phi_{i-1} \\ &= \text{Total cost} + \underbrace{\phi_n}_{\text{final}} - \underbrace{\phi_0}_{\text{Initial}} \end{aligned}$$

$$\Rightarrow \text{Total cost} = \text{Total Amortised cost} - (\phi - \phi)$$

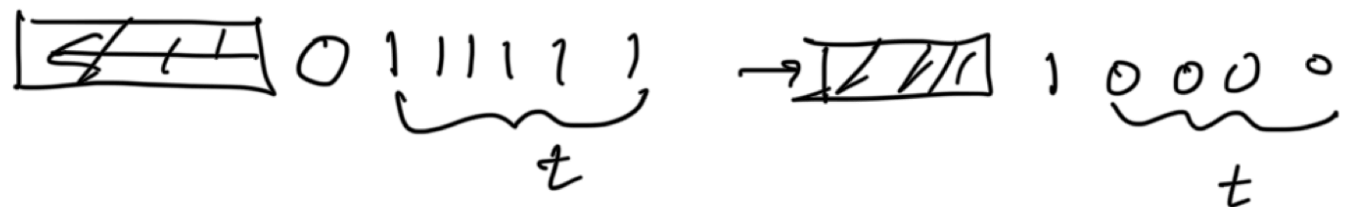
Revisiting the analysis of the n bit counter

$$\phi(\text{count}) = \# 1^s \text{ in the representation}$$

$$\phi(0) = 0 \quad \phi(2^n - 1) = n$$

$$\phi(k+1) - \phi(k) : \text{change in potential going from } k \rightarrow k+1$$

Suppose - there is an initial block of t 1^s



$$\Delta(\phi) = -t + 1$$

Cost of incrementing the count = # bits that change

$$\text{So amortized cost} = -t+1 + t+1 = 2$$

$$\Rightarrow \text{Total amortized cost} = \# \text{ steps} \times 2$$

$$= (2^n - 1) \times 2$$

$$\sim 2 \cdot 2^n$$

We have a number of subsets of a given set S (of n elements)

$S_1, S_2, S_3, \dots, S_k$

Operations of interest : Search / Find

1. Find(x) : return the set(s)
 $\in S$ that contain x

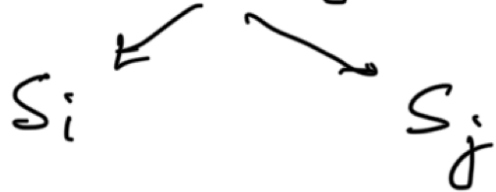
Assumption: 1. $\bigcup S_i = S$

2. $S_i \cap S_j = \emptyset$ i.e. disjoint
 i.e. S_i 's form a partition of S

$$S = \{e_1, e_2, \dots, e_n\}$$

— $S_1 = \{e_1\}$ $S_2 = \{e_2\}$... $S_n = \{e_n\}$
extreme cases $S_1 = S$: trivial because there is only set
 $\text{Find}(e_i) = i$

2. Union operation $S_i \cup S_j$



Case 1 : The subsets are maintained
 as "stars"



e_{i_1} e_{i_2} $\swarrow$ e_{i_k}

$\text{Find}(e) =$ return the label of its
 parent (root) node
 $O(1)$

Union $S_i \cup S_j = S_k$

Change $\min \{ |S_i|, |S_j| \}$ elements

Initially

$$S_i = \{e_i\}$$

We will do a sequence of $N \leq n-1$ unions
 and m Find operations

What is the total cost

$$= O(m + n \cdot N)$$

For each element e_i , we have a hypothetical count c_i that will keep track of the # union operations e_i has participated and its label has changed

$$\Rightarrow O\left(\sum_{i=1}^n c_i\right) = \text{total cost of all unions} \\ \leq n \cdot \left\{ \begin{array}{l} \text{max change in} \\ \text{a counter} \end{array} \right\}$$

$$\max c_i \leq \log_2 n \quad n \cdot \max_i c_i$$

Since the subset that contains an element at least doubles after the label changes

Total : $O(m + n \log n)$

To think : Can we do similar analysis
by defining an appropriate ϕ